1st UG CAPSTONE PROJECT STATUS CHECKING (Week 5)

Project Title:	**Evidence-based provenance scoring for creative digital media content**
Project Client:	Alt.VFX
Group Tag:	COMP3888_M17_01_ZL

Project Information	**Summary of report & feedback**
Overall Information	
Brief on your project requirements, scope, and your proposed computational solutions based on Group's understanding	Project objective and scope. Assess whether submitted creative-process evidence forms a credible, internally consistent and economically plausible account of how a final recording was produced. The selected domain is recorded music/audio. The system is not a binary AI detector, and missing evidence is reported as not assessed or insufficient evidence rather than treated as suspicious. Evidence model. A versioned generic JSON bundle records the final artefact, supporting artefacts, content hashes and declared, inferred or verified relationships. A separate Audio Evidence Profile documents production stages, evidence classes, applicability, relative weights, forge-cost rationale and validation rules. Proposed computational solution. Discovery -> format-specific parsing -> normalisation and classification -> relationship extraction -> artefact and cross-evidence validation -> multidimensional scoring -> human-readable report. The output keeps completeness, integrity and attestation strength separate, each with its own confidence or sufficiency state. C2PA is validated when present, while DDEX RIN and ERN supply session-credit and disclosure semantics.
	Client's feedback N/A
Project milestones and member's work splits and responsibilities	Milestone: Build and integrate parsers, normalisers, bundle construction, relationship resolution and validators. Implement scoring system; demonstrate C2PA interoperability; test development, unseen and adversarial bundles; report false positives and confidence intervals. Recorded work split: Hongjie - relationship resolution and validator and Building scoring system; William - WAV parser for the Cambridge datasets; Declan - CueSheet/CompSheet PDF and text parser; Mars - LMMS parser; Nick and Yee - C2PA/ERN/RIN parsers; Danny and Peter - bounded DAW investigation. Each parser owner also normalises outputs into the common artefact format; schema coordination, integration, testing and documentation are shared responsibilities.

<table>
<tr><td></td><td>Client’s feedback
N/A</td></tr>
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<tr><th>Status Highlight</th><th>Summary of Project Progress</th></tr>
<tr><td>Overall Project Status</td><td>Design established; implementation in progress. The scope, evidence classes, candidate datasets and processing pipeline are defined. Schema/profile work is at design level, while end-to-end ingestion, validation, scoring and evaluation are still under development.</td></tr>
<tr><td rowspan="2">Progress and Achievements</td><td>Scope and dataset investigation
• The team narrowed several candidate media domains to recorded music/audio and mapped evidence across ideation, recording, selection/editing, mixing, mastering and the final track.
• Cambridge-MT Mixing Secrets, Cambridge-MT Recording Secrets, TELEFUNKEN multitracks and the LMMS Sharing Platform were investigated for multitracks, takes/comps, production documents, stems, mixes, masters and project-file evidence.

Schema, relationships and architecture
• The three-part design separates the generic Evidence Bundle Schema, Audio Evidence Profile and common Score Vector. Original files remain separate and are linked by stable identifiers, SHA-256 hashes and an evidence graph leading toward the final recording.
• Relationships such as selected_into, grouped_into, mixed_into, derived_from, documents/describes and references were identified. Every relationship retains its declared, inferred or verified basis, confidence and rationale.</td></tr>
<tr><td>Client’s feedback:
The client is happy with our current progress and says that we are on the right track with our schema and pipeline.</td></tr>
<tr><td>Key Issues</td><td>The main issues are shared-schema alignment, heterogeneous inputs, standards and the risk of false certainty from weak evidence.</td></tr>
<tr><td rowspan="2">Obstacles & Risks</td><td>1.Schema versions and parser outputs must stay compatible with the current team repository and the cross-group test harness; design-level prototypes still need integration and real-dataset verification.
2.Mixed formats, malformed files, missing metadata and weak timestamps can produce false certainty. Unknown values must remain unknown and must not be converted to zero or labelled inconsistent.
3.Lacking DDEX&&C2PA-related dataset</td></tr>
<tr><td>Client’s feedback:
The client said that it would be looked upon highly to have a good amount of compatibility between domains, so encourages as much collaboration as possible. Also that incomplete or malformed input is a big part of the challenge of this project and to make sure our system can handle that.</td></tr>
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Individual Engagement & Contribution (up to 0.5 page / member)

Hongjie Deng	
Participation and attendance of meetings	client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	My primary responsibilities included coordinating task allocation, designing the project workflow, developing the scoring model, and implementing relationship validators. I independently designed and implemented the evidence bundle schema and domain-specific evidence profiles. I also actively collaborated with other teams to align our approaches and contribute to a shared, generalised output schema that supports cross-team integration and interoperability.

shuhaozhang(danny) shuhaozhang	
Participation meetings	client meetings 75%(missing week5) group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	My key role is as a Tester and Programmer, focusing on system validation and investigating edge-case evidence classes. My primary work split involves leading the bounded DAW investigation, where I assess the feasibility, structural limitations, and evidentiary value of proprietary DAW project files as a source of provenance. My most profound contribution has been identifying realistic bounded DAW scenarios—where critical metadata is missing, overwritten, or obfuscated—and proposing clear mitigation strategies to handle these gaps without introducing false certainty into our scoring pipeline.

Khanh Phan (Peter)	
Participation meetings	client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	My key roles in the project include developing presentations, providing administrative support, and research. I designed the project slides to express our core ideas and approach, assisting with tasks when needed. My primary technical contributions are on researching audio production evidence collection and verification, focusing on Digital Audio Workstation (DAW), WAV audio, and production records. I examined methods to extract file properties using Python into a group made schema, which is used as evidence collection for our human logging and verification system.

Declan Schulz	

Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	My key roles were initially research on various evidence classes, their structure and function within the music production workflow. Since then I have focused on DDEX (primarily RIN) as well as cue and comp sheets, particularly understanding the fields that are useful and how they will contribute as evidence in the final score (alone and in a relationship to evidence sources). My largest contribution is as above, developing the logic that will form the basis of our scoring system.

Nick Chen	
Participation meetings	Participation of client meetings 100% Group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	My key roles and responsibilities were to initially look into subdomains we would be interested in wherein we decided music. From here, it has been mostly to begin work on helping with parsers for ERN and RIN. From here it has been to continue work on how to normalise this afterwards. Moreover, it has been to look at how we want our overall architecture to look including schemas, specifically how integration would look with other teams and whether design choices such as a DAG traversal for combining Vector scores is suitable for us. I have also kept meeting minutes from week 2 onwards.

William Alexander	
Participation meetings	Participation of client meetings 100% group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	My key roles were to first research potential evidence sources, primarily the Cambridge-MT recording secrets dataset. I then worked primarily on a parser for WAV files. To complete this, I stripped any available metadata from the file and output this into JSON format in a portable python package. In addition to JSON data stripping, I am in the planning stage to determine how best to use the raw audio data from WAV files in the evidence scoring process. I also kept meeting minutes for week 4.

Marsya Amanda	
Participation meetings	Participation of client meetings 100% group meetings 80% (missing in W5 in- and out-of-tutorial group meetings)
Major roles, work split, & highlight the most profound contributions to the project progress	My key roles were to communicate team progress and concerns to the client during client meetings, research and collect findings on music production workflows with and without AI-assistance, as well as explore the stems evidence class through ad-hoc literature review and utilising computational tools. I recorded meeting minutes in week 1. I am in the process of finding a dataset of DAW (process) files.

Yee Cheng Ng yee

Participation meetings	Participation of client meetings 75% (missing meeting in week 5), group meetings 100%
Major roles, work split, & highlight the most profound contributions to the project progress	- Developed baseline parsers for C2PA, DDEX RIN, and DDEX ERN to support extraction and standardisation of provenance evidence. - Conducted TELEFUNKEN multitrack audio analysis, exploring RMS-based cross-evidence relationships and consistency checking for audio provenance. - Led and maintained internal project documentation